

AI Assistant Development

VaultofCodes College Project

Developer & Creator: Chhotelal Kushwaha

GitHub Repository Link:

<https://github.com/karan2027/VOC-AI-Assistant-Development>

LinkedIn Activity Post Link:

https://www.linkedin.com/posts/chhotelal-kushwaha-2902a3329_vaultofcodes-promptengineering-artificialintelligence-activity-7486847656430940160-oi-f?utm_source=share&utm_medium=member_android&rcm;=ACoAAFLNZ8wBTSjgCxbdn5SefSpljH7s0IzWm0k

Project Details & Summary

VaultofCodes AI Assistant is a full-stack, web-based Artificial Intelligence application engineered to deliver intelligent conversational capabilities, dynamic text summarization, creative content generation, and tailored step-by-step advice.

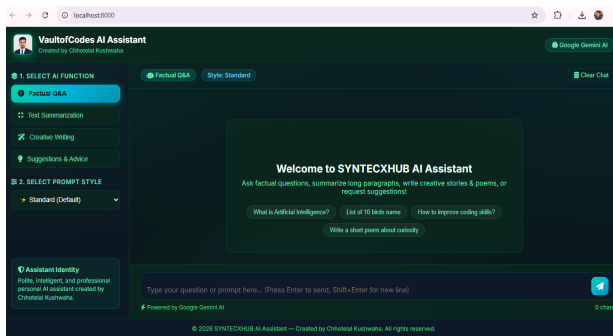
Core Capabilities & Requirements Fulfilled:

- **Factual Q&A;**: Provides concise and accurate answers to general knowledge and technical questions.
- **Text Summarization**: Extracts executive summaries and structured bullet points from long paragraphs.
 - **Creative Content Generation**: Writes imaginative stories, poems, speeches, and creative texts.
- **Suggestions & Advice**: Generates practical, structured coaching and actionable recommendations.

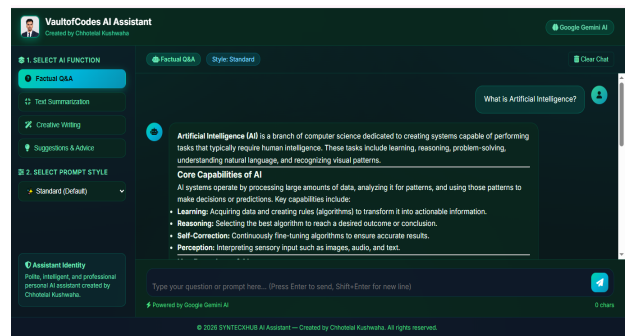
Technical Architecture & Stack:

- **Backend Framework**: Python 3 & Flask REST API Service.
- **AI Core Engine**: Google Gemini API (v1beta REST) with Multi-Provider Fallback.
- **Frontend Interface**: Responsive HTML5, Vanilla CSS3 (Dark Glassmorphic Theme), JavaScript (ES6+).

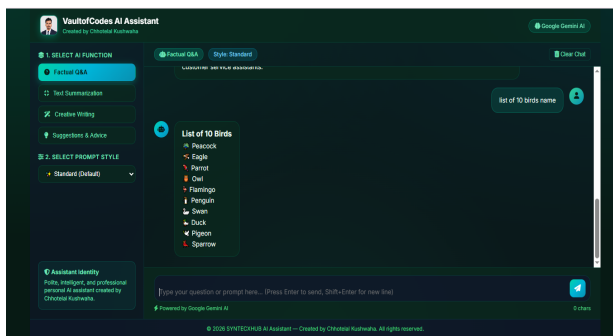
Project Screenshots Overview



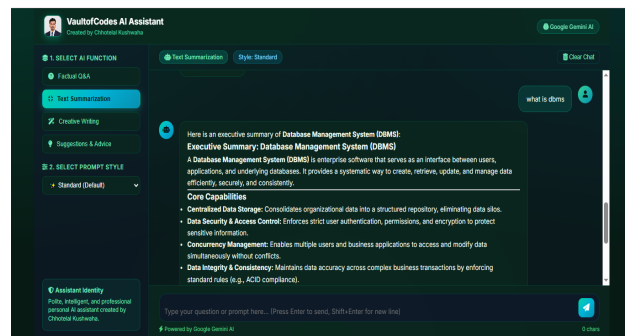
1. Main UI (Home)



2. Factual Q&A;

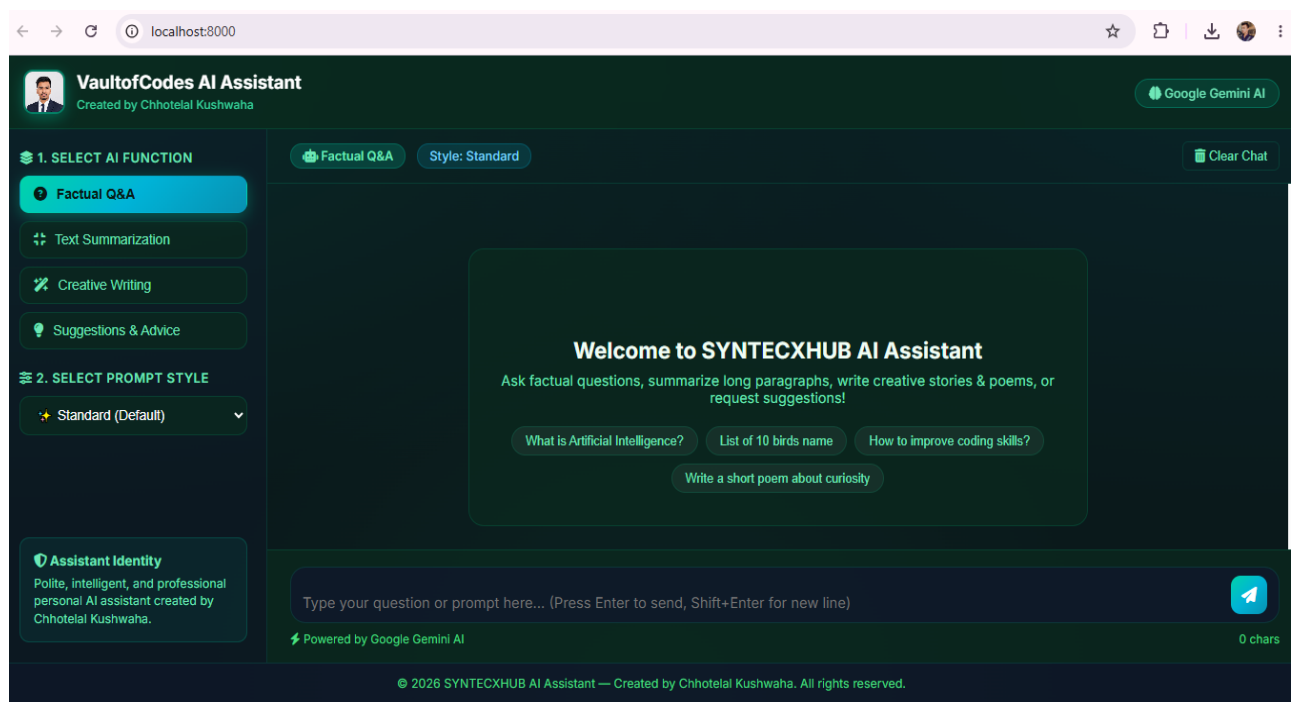


3. List Generation



4. Text Summarization

Output 1: Main Application Interface (Home View)



Output 2: Factual Q&A; Mode — Artificial Intelligence Query

The screenshot displays the VaultofCodes AI Assistant interface. At the top, the header shows the assistant's name 'VaultofCodes AI Assistant' and its creator 'Created by Chhotelal Kushwaha'. A 'Google Gemini AI' badge is visible in the top right corner. The interface is divided into a left sidebar and a main chat area.

Left Sidebar:

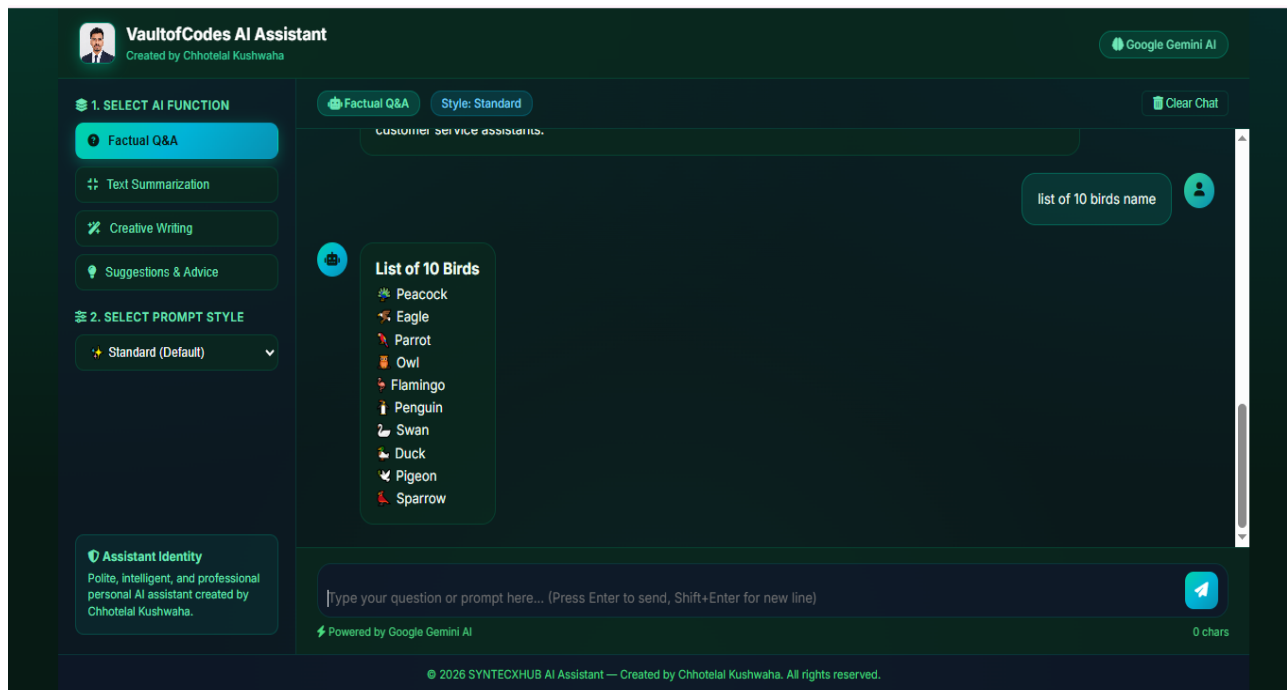
- 1. SELECT AI FUNCTION:** Includes buttons for 'Factual Q&A' (selected), 'Text Summarization', 'Creative Writing', and 'Suggestions & Advice'.
- 2. SELECT PROMPT STYLE:** A dropdown menu showing 'Standard (Default)'.
- Assistant Identity:** A section describing the assistant as 'Polite, intelligent, and professional personal AI assistant created by Chhotelal Kushwaha'.

Main Chat Area:

- Buttons for 'Factual Q&A' and 'Style: Standard' are at the top.
- A 'Clear Chat' button is in the top right.
- The user's query 'What is Artificial Intelligence?' is shown in a blue bubble.
- The assistant's response is in a white bubble, starting with a robot icon. The text reads: 'Artificial Intelligence (AI) is a branch of computer science dedicated to creating systems capable of performing tasks that typically require human intelligence. These tasks include learning, reasoning, problem-solving, understanding natural language, and recognizing visual patterns.' This is followed by a section titled 'Core Capabilities of AI' which explains that AI systems operate by processing large amounts of data and lists four key capabilities: Learning, Reasoning, Self-Correction, and Perception.
- At the bottom, there is a text input field with the placeholder 'Type your question or prompt here... (Press Enter to send, Shift+Enter for new line)' and a blue send button.
- Below the input field, it says 'Powered by Google Gemini AI' and '0 chars'.

Footer: © 2026 SYNTECXHUB AI Assistant — Created by Chhotelal Kushwaha. All rights reserved.

Output 3: Data Formatting & List Generation (10 Birds Name)



Output 4: Text Summarization Mode — DBMS Executive Summary

The screenshot displays the VaultofCodes AI Assistant interface. At the top, the assistant is identified as 'VaultofCodes AI Assistant' created by 'Chhotelal Kushwaha', powered by 'Google Gemini AI'. The interface is divided into a left sidebar and a main chat area.

Left Sidebar:

- 1. SELECT AI FUNCTION:** Includes buttons for 'Factual Q&A', 'Text Summarization' (highlighted in blue), 'Creative Writing', and 'Suggestions & Advice'.
- 2. SELECT PROMPT STYLE:** A dropdown menu showing 'Standard (Default)'.
- Assistant Identity:** A box stating 'Polite, intelligent, and professional personal AI assistant created by Chhotelal Kushwaha.'

Main Chat Area:

- Buttons for 'Text Summarization' and 'Style: Standard' are at the top.
- A 'Clear Chat' button is in the top right.
- The user's prompt 'what is dbms' is shown in a blue bubble.
- The assistant's response is in a green bubble, starting with 'Here is an executive summary of Database Management System (DBMS):'.
- Executive Summary: Database Management System (DBMS)**
A Database Management System (DBMS) is enterprise software that serves as an interface between users, applications, and underlying databases. It provides a systematic way to create, retrieve, update, and manage data efficiently, securely, and consistently.
- Core Capabilities**
 - **Centralized Data Storage:** Consolidates organizational data into a structured repository, eliminating data silos.
 - **Data Security & Access Control:** Enforces strict user authentication, permissions, and encryption to protect sensitive information.
 - **Concurrency Management:** Enables multiple users and business applications to access and modify data simultaneously without conflicts.
 - **Data Integrity & Consistency:** Maintains data accuracy across complex business transactions by enforcing standard rules (e.g., ACID compliance).
- A text input field at the bottom says 'Type your question or prompt here... (Press Enter to send, Shift+Enter for new line)'.
- A status bar at the bottom indicates 'Powered by Google Gemini AI' and '0 chars'.